AWS VPC Peering Project

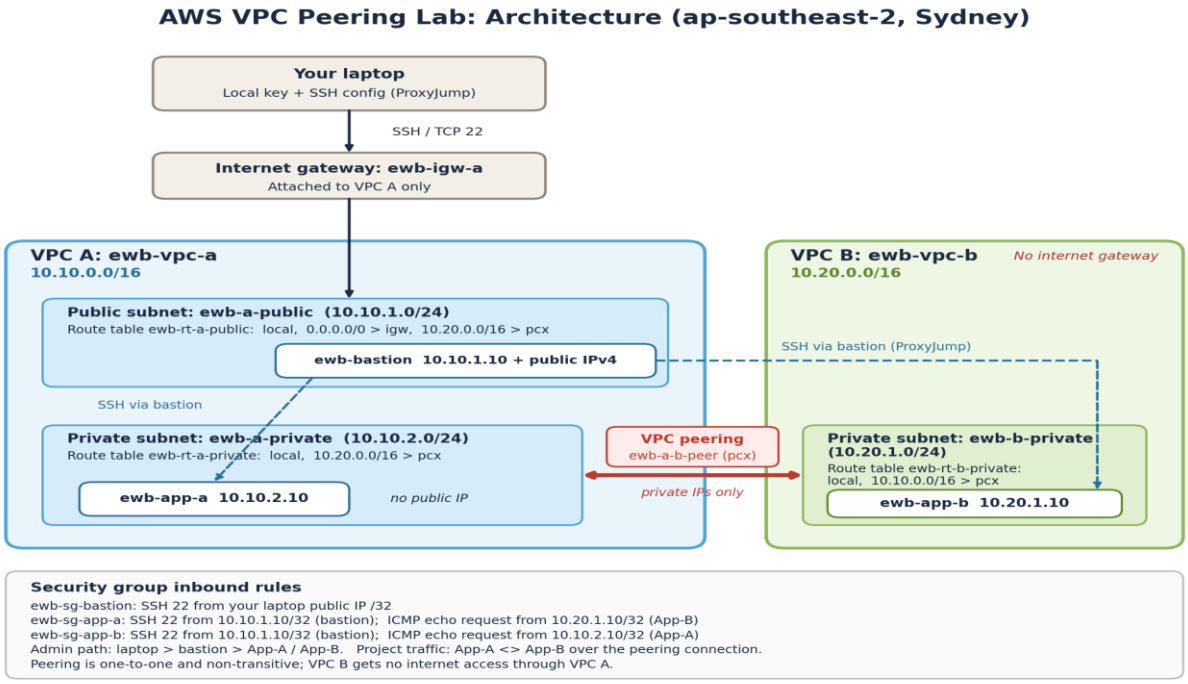
Name: Harshini Nirmalkumar
Batch: Batch 1, AWS Cloud Computing Internship (EWB Courses)
Project: Project 01, VPC Peering
Region: Asia Pacific (Sydney), ap-southeast-2, Availability Zone ap-southeast-2a
Completion date: 19 October 2026
GitHub repository: https://github.com/harshini346/aws-vpc-peering-lab

1. Objective

The goal was to connect two VPCs with VPC peering so that two private EC2 servers can exchange traffic over their private IPv4 addresses, while keeping both servers private. One public bastion host in VPC A is the only administration entry point. I built the lab, proved it failed before peering, proved it worked after peering, and ran a controlled failure to show the role of the security group rule.

The lab was built with AWS CLI scripts run in AWS CloudShell (see the GitHub repository) and verified from my laptop over SSH using ProxyJump through the bastion.

2. Architecture and resource inventory



Traffic paths

- Administration: laptop, then bastion, then App-A or App-B using SSH ProxyJump.
- Project traffic: App-A and App-B talk over private IPs through the peering connection.

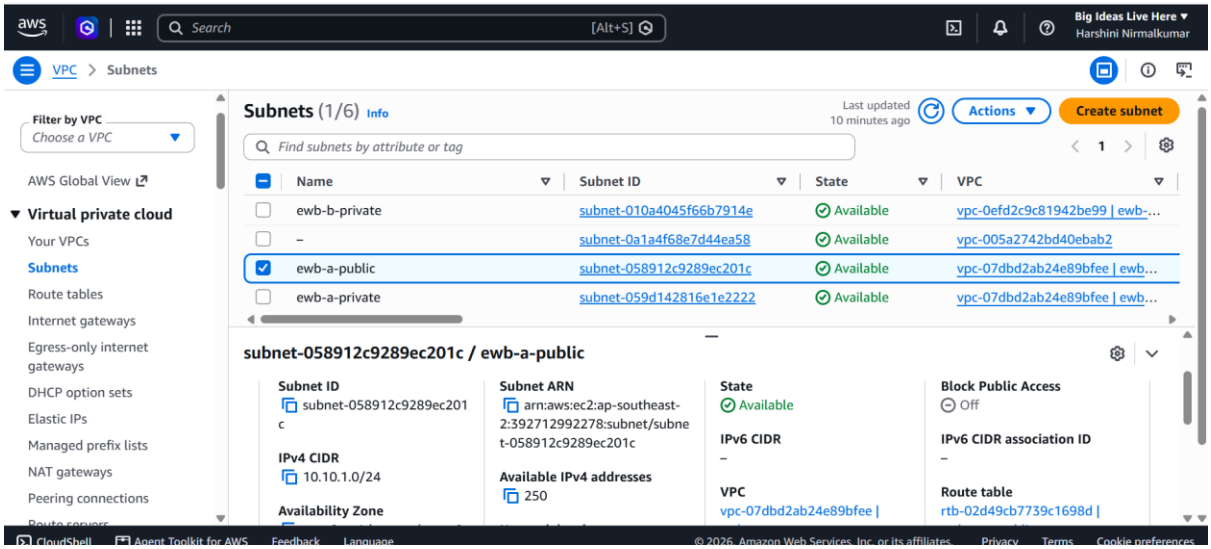
- VPC B has no internet gateway. Peering does not give VPC B internet access through VPC A.

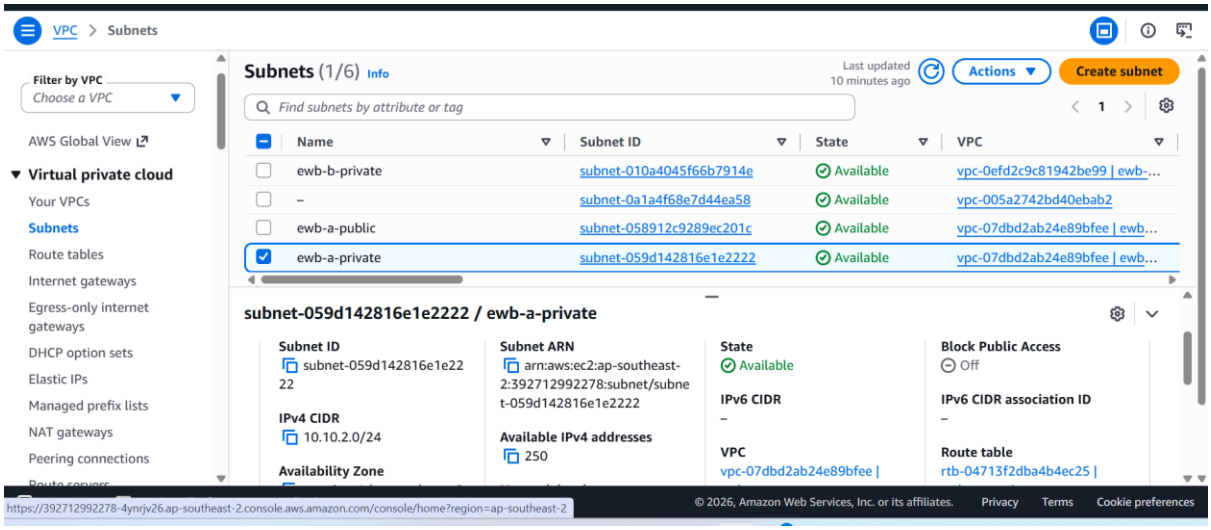
Resource	Name	CIDR / address	ID
VPC A	ewb-vpc-a	10.10.0.0/16	vpc-07dbd2ab24e89bfee
Public subnet A	ewb-a-public	10.10.1.0/24	subnet-058912c9289ec201c
Private subnet A	ewb-a-private	10.10.2.0/24	subnet-059d142816e1e2222
VPC B	ewb-vpc-b	10.20.0.0/16	vpc-0efd2c9c81942be99
Private subnet B	ewb-b-private	10.20.1.0/24	subnet-010a4045f66b7914e
Internet gateway	ewb-igw-a	attached to VPC A	igw-096079d859cfd8cdd
Peering connection	ewb-a-b-peer	requester VPC A, accepter VPC B	pcx-08c41d6b2ffbeb11e
Bastion	ewb-bastion	10.10.1.10 + public IPv4 3.106.122.133	i-073f0633fae84dc2d
App-A	ewb-app-a	10.10.2.10, no public IP	i-026276b5ced8293b3
App-B	ewb-app-b	10.20.1.10, no public IP	i-096a5c483ce4b6565

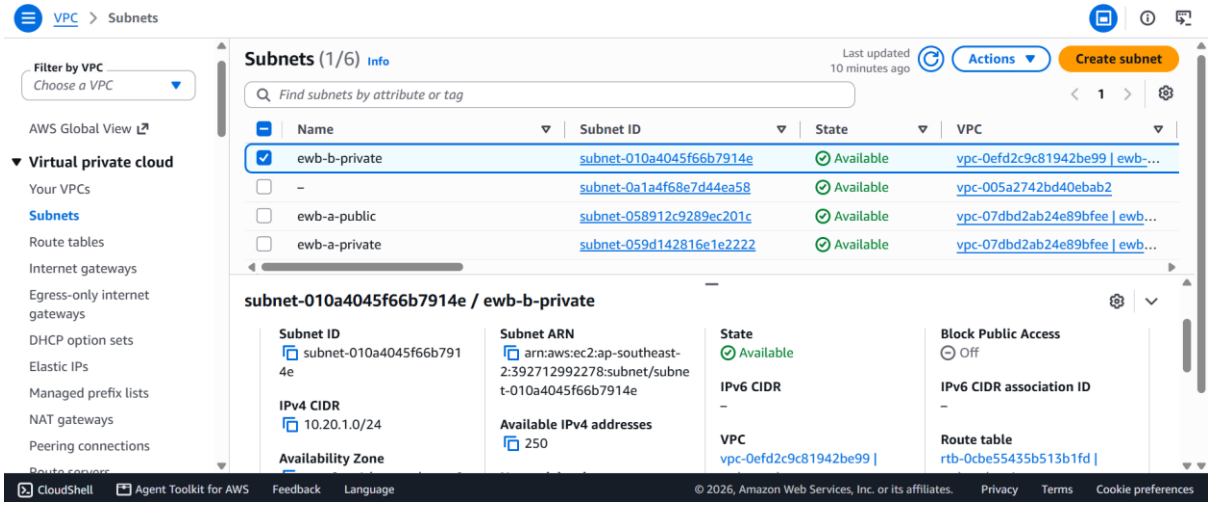
All resources were tagged Project = EWB-VPC-Peering and Student = Harshini. All instances are Amazon Linux 2023 on t3.micro with an 8 GB root volume set to delete on termination.

Why these CIDRs: 10.10.0.0/16 and 10.20.0.0/16 do not overlap, which is required for peering. Different subnet ranges alone would not fix overlapping VPC CIDRs.

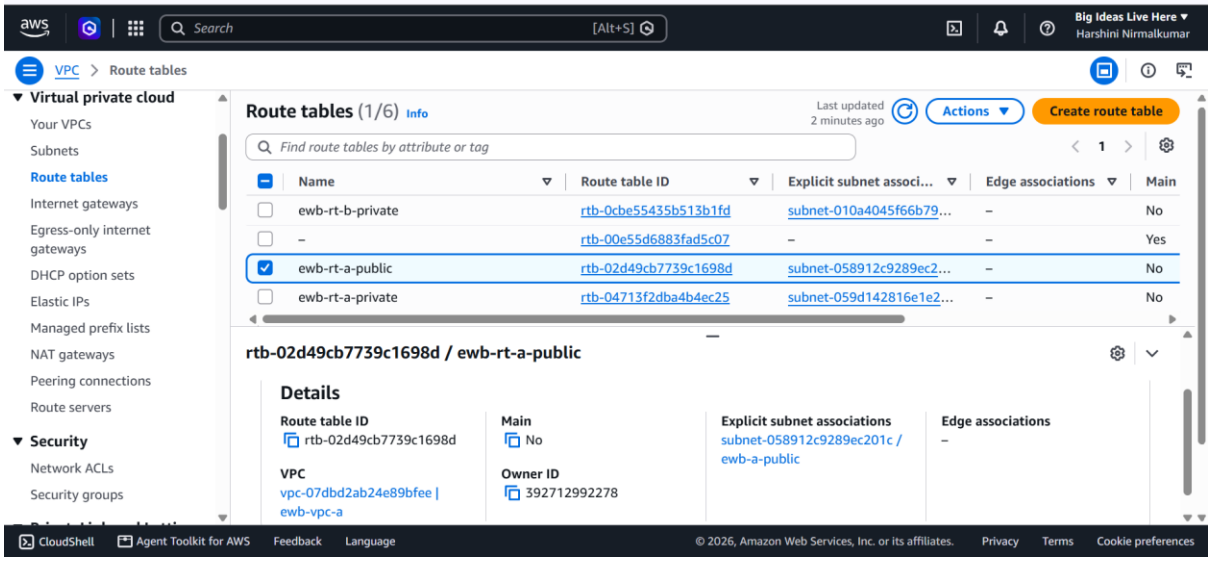
3. Subnets, internet gateway and route tables

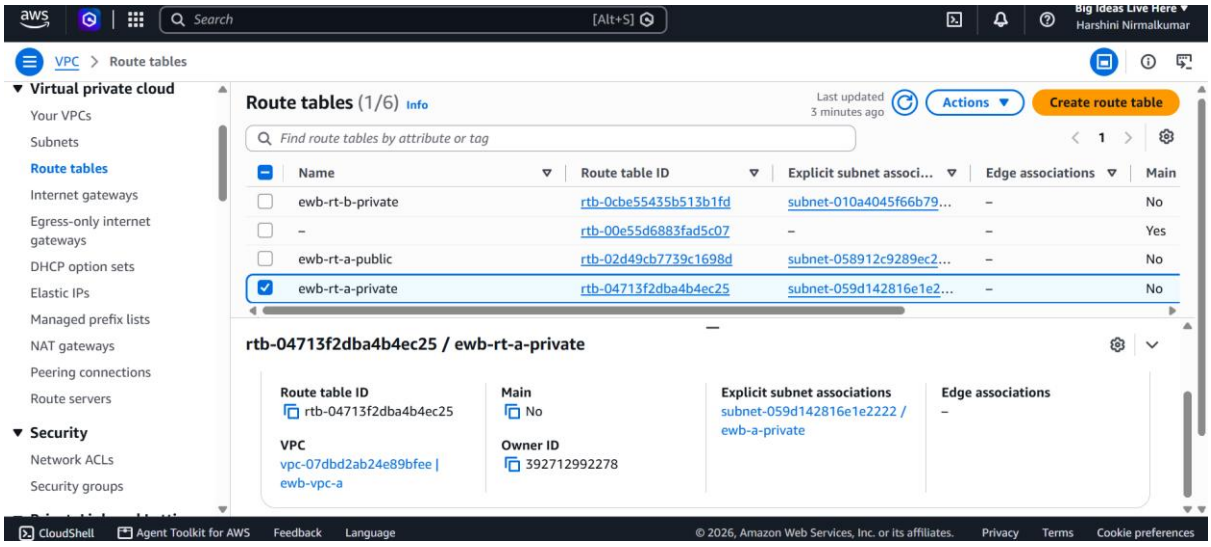


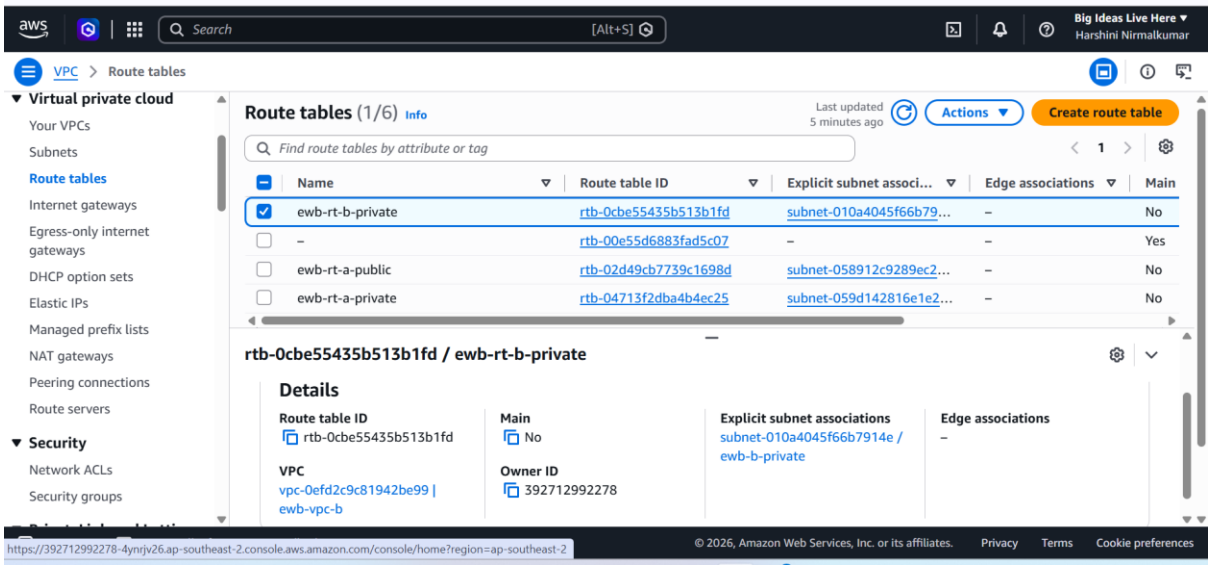




Route table	Associated subnet	Association ID	Routes
ewb-rt-a-public (rtb-02d49cb7739c1698d)	ewb-a-public	rtbassoc-0a244bfefe346fe41	10.10.0.0/16 local; 0.0.0.0/0 to ewb-igw-a; 10.20.0.0/16 to peering
ewb-rt-a-private (rtb-04713f2dba4b4ec25)	ewb-a-private	rtbassoc-00c8c03f9607c645e	10.10.0.0/16 local; 10.20.0.0/16 to peering
ewb-rt-b-private (rtb-0cbe55435b513b1fd)	ewb-b-private	rtbassoc-0ec35e6bef6ab66d8	10.20.0.0/16 local; 10.10.0.0/16 to peering



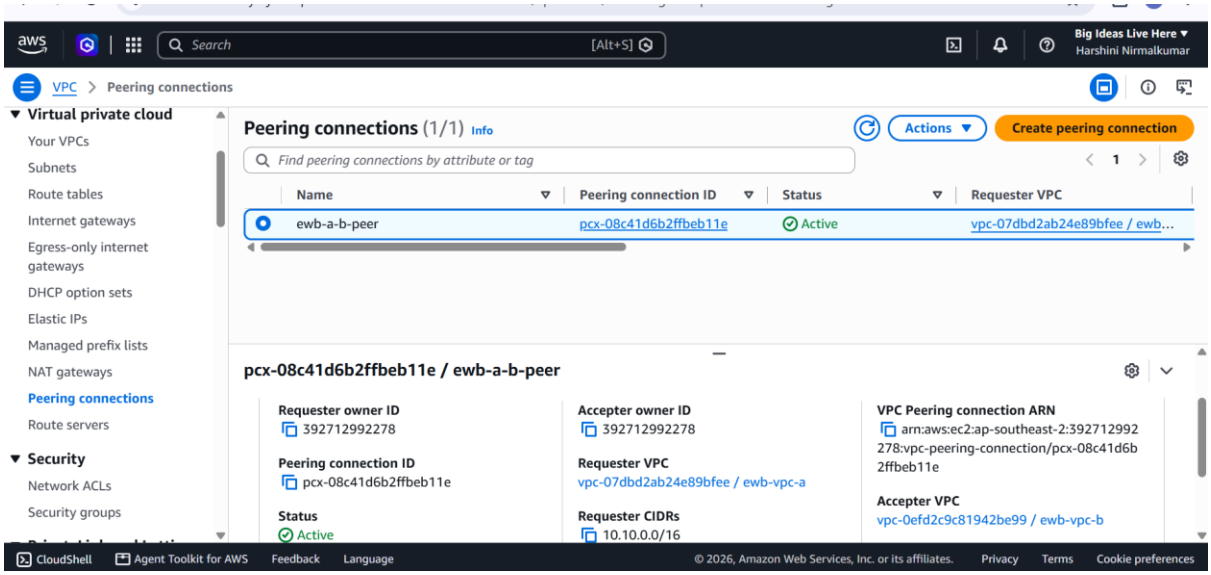




The private route tables have no internet default route, and auto-assign public IPv4 is disabled on the private subnets. Only the bastion received a public IPv4 address.

4. Peering connection

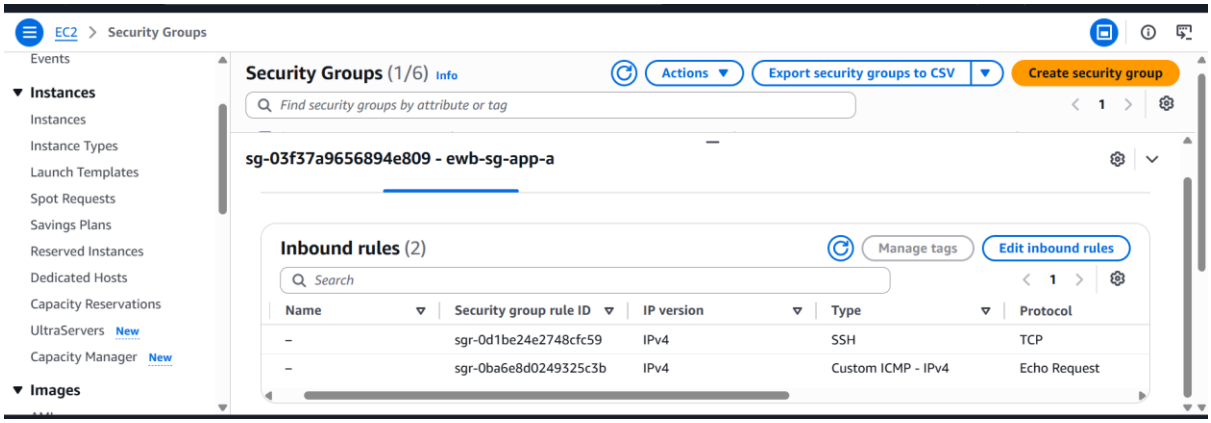
The peering connection ewb-a-b-peer (pcx-08c41d6b2ffbeb11e) was requested from VPC A (requester) to VPC B (accepter) in the same account and Region. A request must be accepted even when both VPCs are in one account; I accepted it and waited for the status to become Active.

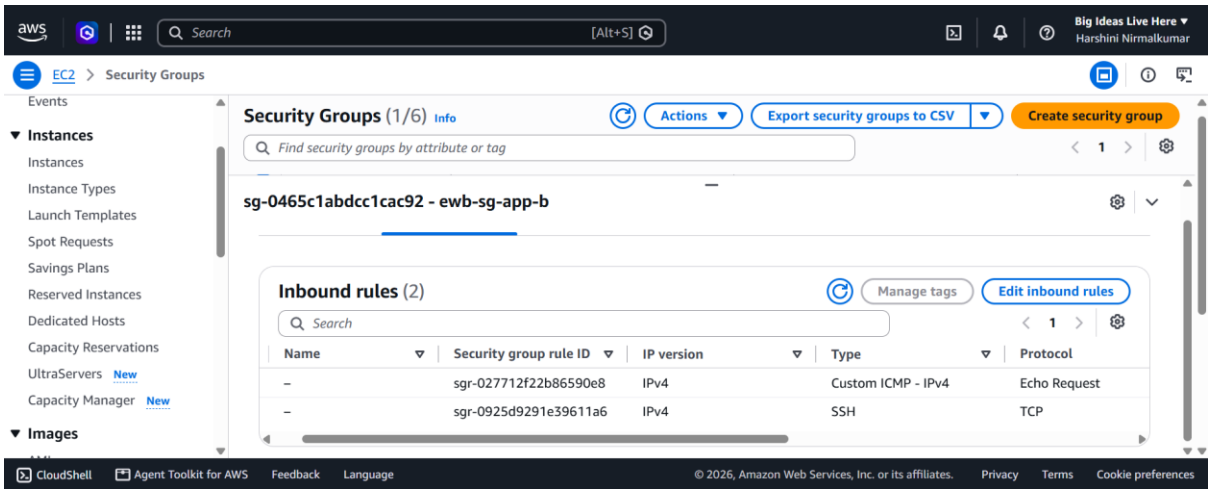


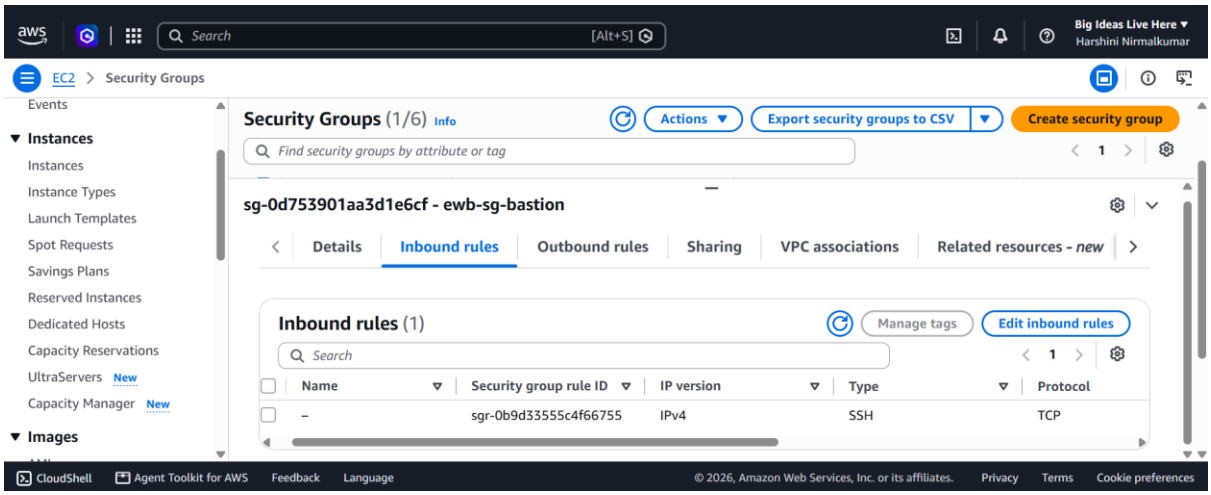
5. Security group rules

Security group	Inbound rule	Source
ewb-sg-bastion (sg-0d753901aa3d1e6cf)	SSH, TCP 22	my laptop public IPv4 /32
ewb-sg-app-a (sg-03f37a9656894e809)	SSH, TCP 22	10.10.1.10/32 (bastion)
ewb-sg-app-a	ICMP echo request (type 8, code 0)	10.20.1.10/32 (App-B)
ewb-sg-app-b (sg-0465c1abdcc1cac92)	SSH, TCP 22	10.10.1.10/32 (bastion)
ewb-sg-app-b	ICMP echo request (type 8, code 0)	10.10.2.10/32 (App-A)

Outbound rules were left as the default allow-all, and network ACLs were the default allow-all. Security groups are stateful, so replies to permitted traffic are allowed automatically. SSH and ICMP were never opened to 0.0.0.0/0.

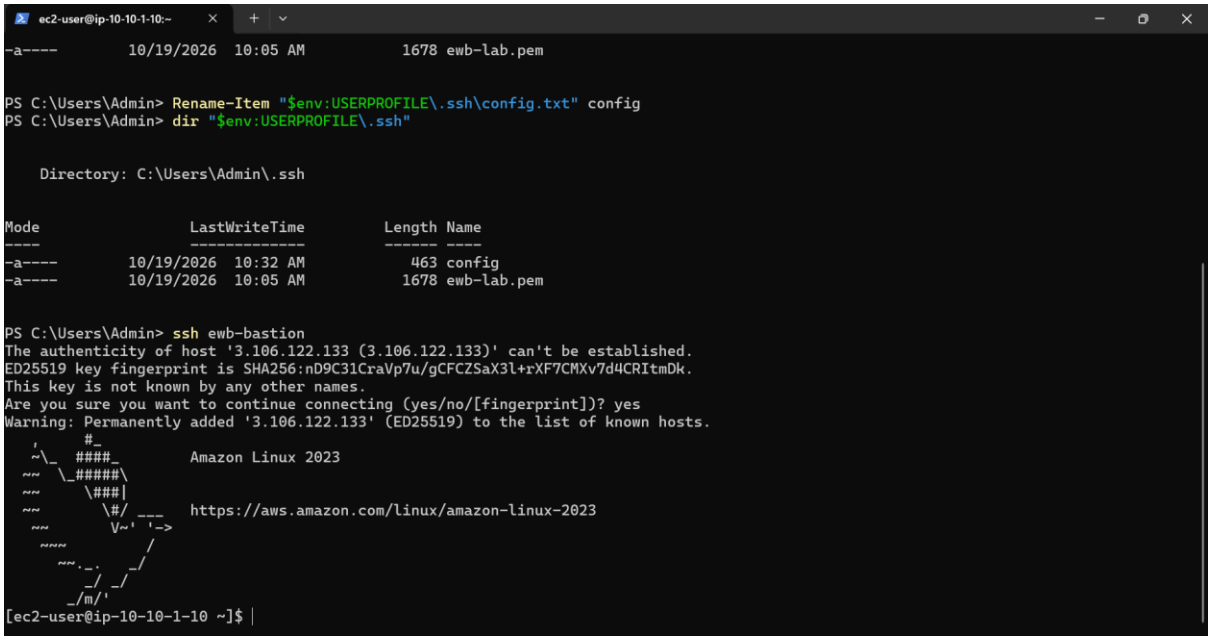






6. Private SSH access

I stored the key pair ewb-lab only on my laptop and never copied it to any instance. SSH uses ProxyJump through the bastion. I connected to the bastion first, then to App-A. After peering and routes were in place, I connected to App-B through the bastion as well.



7. Test results

Test	Where / command	Expected	Result
01 Baseline (before peering)	App-A: ping -c 4 10.20.1.10	No replies	4 transmitted, 0 received, 100% packet loss
02 Forward path (after routes)	App-A: ping -c 4 10.20.1.10	4 replies	4 transmitted, 4 received, 0% packet loss
03 Reverse path (after routes)	App-B: ping -c 4 10.10.2.10	4 replies	4 transmitted, 4 received, 0% packet loss
04 Private SSH	Laptop: ssh ewb-app-b	Shell on App-B via bastion	Connected, prompt ip-10-20-1-10
05 Isolation	EC2 instance details	No public IPv4 on App-A and App-B	Public IPv4 field empty on both

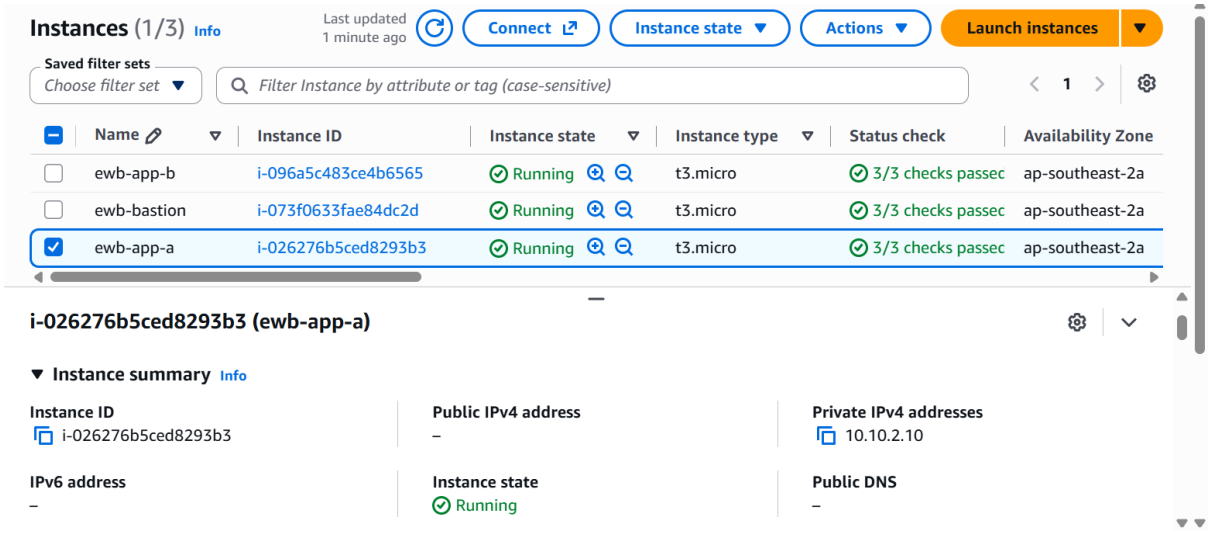


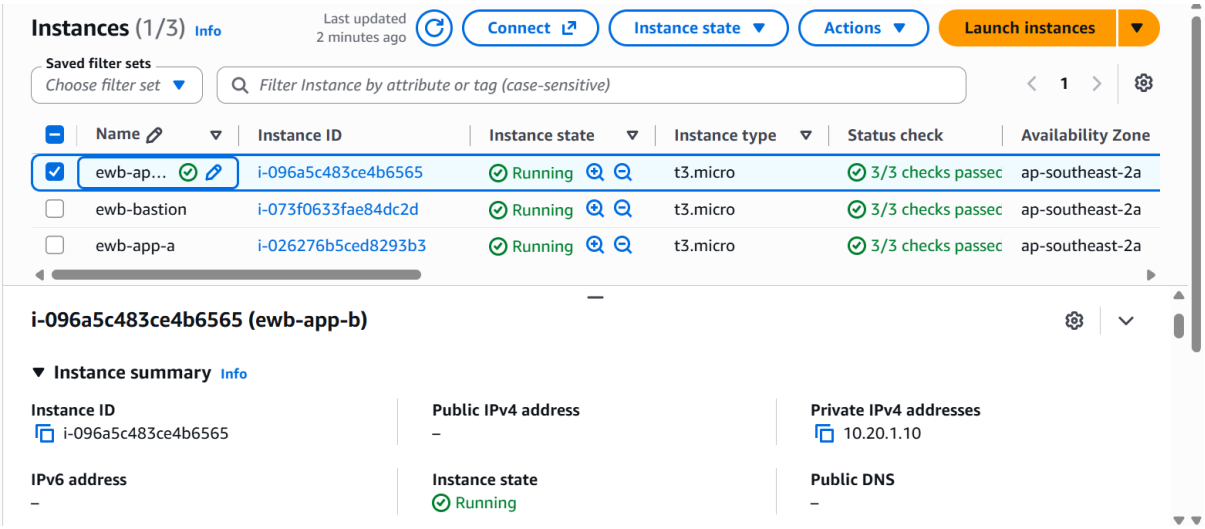
```
Connection to 10.10.2.10 closed.
PS C:\Users\Admin> ssh ewb-app-a\
Hostname contains invalid characters
PS C:\Users\Admin> ssh ewb-app-a
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Mon Oct 19 05:07:23 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.499 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.188 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.187 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.197 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3111ms
rtt min/avg/max/mdev = 0.187/0.267/0.499/0.133 ms
[ec2-user@ip-10-10-2-10 ~]$ ssh ewb-app-b
ssh: Could not resolve hostname ewb-app-b: Name or service not known
[ec2-user@ip-10-10-2-10 ~]$ exit
logout
Connection to 10.10.2.10 closed.
PS C:\Users\Admin>
```

```
Connection to 10.10.2.10 closed.
PS C:\Users\Admin> ssh ewb-app-b
The authenticity of host '10.20.1.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:exHBCDE5KQeSHHHmpAy/dsAipeRwOWtrdcBR8G1tkQw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.10' (ED25519) to the list of known hosts.
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-20-1-10 ~]$ hostname
ip-10-20-1-10.ap-southeast-2.compute.internal
[ec2-user@ip-10-20-1-10 ~]$ hostname -I
10.20.1.10
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.205 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.191 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.170 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.179 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3135ms
rtt min/avg/max/mdev = 0.170/0.186/0.205/0.013 ms
[ec2-user@ip-10-20-1-10 ~]$
```





8. Controlled failure and recovery

I removed only the inbound ICMP echo request rule (source 10.10.2.10/32) from ewb-sg-app-b and kept the SSH rule. From App-A, ping -c 4 10.20.1.10 then returned 4 transmitted, 0 received, 100% packet loss. I restored the exact same rule and the ping returned 4 replies with 0% packet loss.

This shows that an Active peering connection and correct routes are not enough on their own: the destination security group must also allow the traffic. The route table decides whether a packet can reach the other VPC, and the security group decides whether the destination accepts it.

```
       _/ _/
     _/m/'
ec2-user@ip-10-10-1-10 ~]$ exit
ogout
onnection to 3.106.122.133 closed.
S C:\Users\Admin> ssh ewb-app-a
he authenticity of host '10.10.2.10 (<no hostip for proxy command>)' can't be established.
D25519 key fingerprint is SHA256:vEwtKhK3daup/9FL0m9agg8vn9yeuaULzxKOSFIPIMo.
his key is not known by any other names.
re you sure you want to continue connecting (yes/no/[fingerprint])? yes
arning: Permanently added '10.10.2.10' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
ec2-user@ip-10-10-2-10 ~]$ hostname
p-10-10-2-10.ap-southeast-2.compute.internal
ec2-user@ip-10-10-2-10 ~]$ hostname -I
0.10.2.10
ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
ING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

-- 10.20.1.10 ping statistics ---
 packets transmitted, 0 received, 100% packet loss, time 3127ms
```

9. Forward path and return path

When App-A (10.10.2.10) pings App-B (10.20.1.10), the packet leaves App-A and ewb-rt-a-private matches 10.20.0.0/16 and sends it to the peering connection. It arrives in VPC B, where the security group ewb-sg-app-b allows the echo request from 10.10.2.10/32. App-B's reply is addressed to 10.10.2.10, and ewb-rt-b-private matches 10.10.0.0/16 and sends it back through the peering connection to App-A. If either route is missing, the ping or SSH session fails, because a one-way route is not enough.

10. Questions I should be able to explain

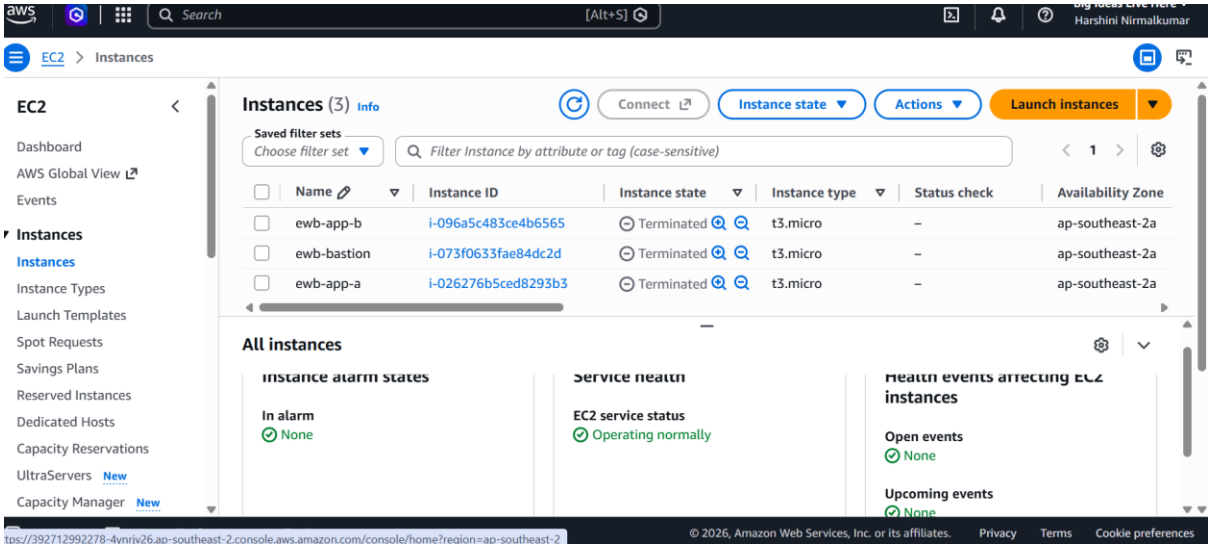
1. **Why is the return route needed?** The reply has to travel back to the other VPC. Without 10.10.0.0/16 pointing at the peering connection in VPC B's route table, App-B has no path to answer App-A.
2. **Why is the App-B SSH source the bastion private IP?** With ProxyJump, the bastion opens a separate SSH connection to App-B, so App-B sees the bastion's private IP 10.10.1.10 as the source, not my laptop.
3. **Why does Active peering not guarantee a successful ping?** Peering is only the link. Route tables must send traffic into it from both sides, and security groups (and network ACLs) must allow the traffic.
4. **Can VPC B use VPC A's internet gateway over peering?** No. Peering is one-to-one and non-transitive, and it does not provide internet access through the other VPC's gateway.

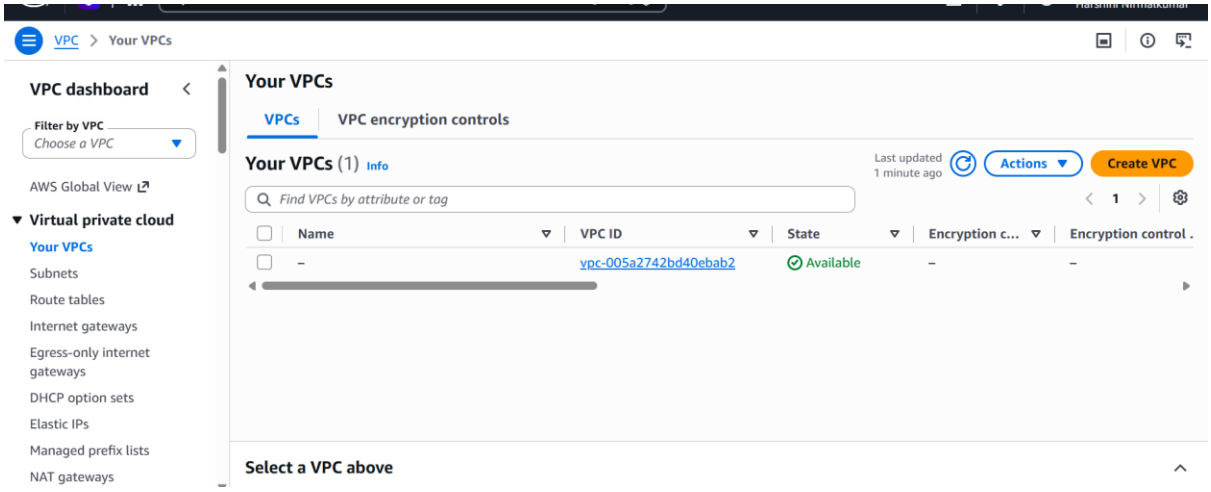
The bastion's public subnet also needs the 10.20.0.0/16 peering route, because the bastion itself opens the connection to App-B. The more specific 10.20.0.0/16 route is chosen over the internet default route.

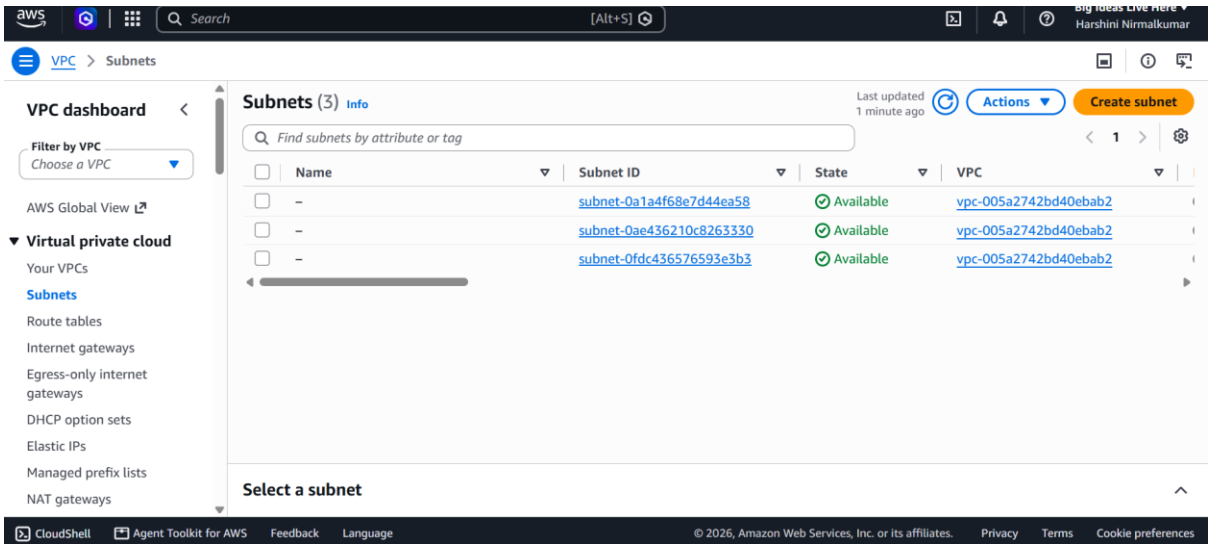
11. Cleanup evidence

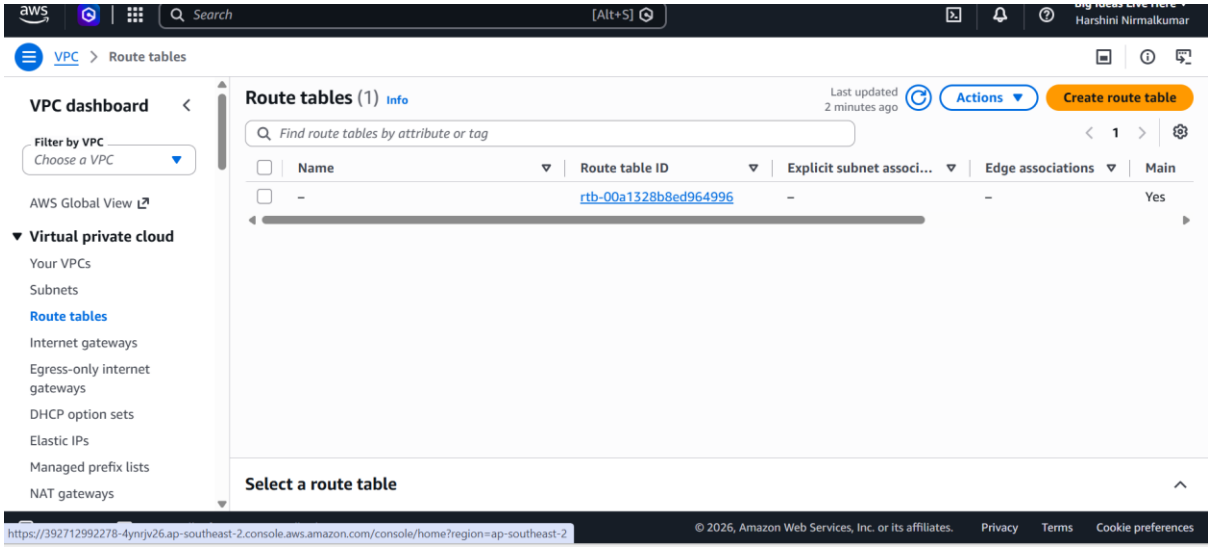
After saving all evidence, I terminated the three instances, deleted the peering routes and the peering connection, the security groups, the route table associations and route tables, the subnets, the internet gateway and both VPCs, and deleted the ewb-lab key pair. I then checked EC2 instances, volumes and the VPC console for leftover resources.

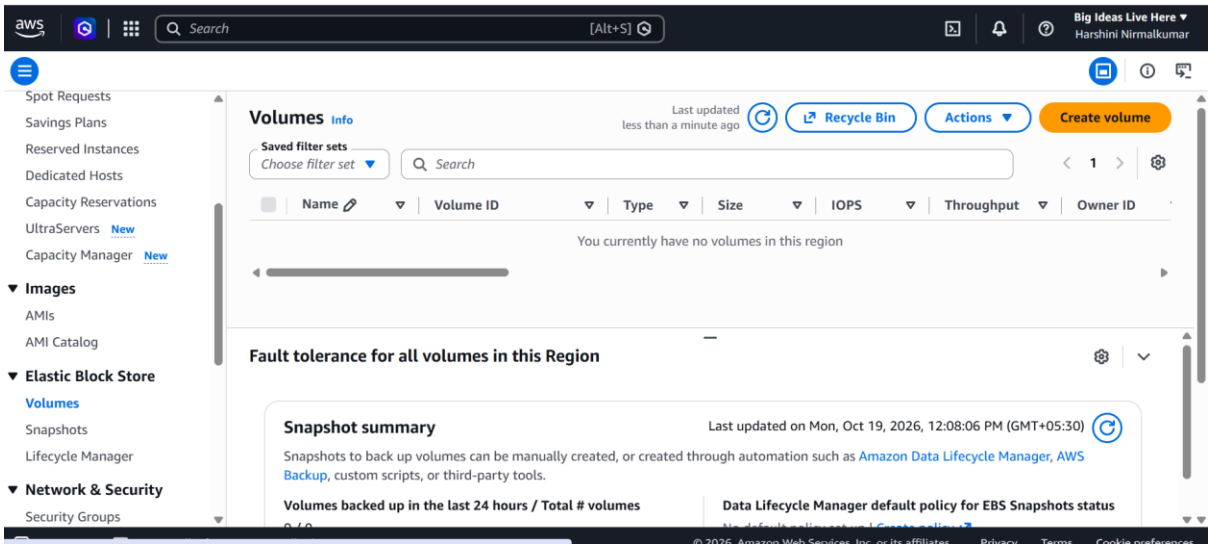



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Terminated
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ap-southeast-2a
All instances
In alarm
None
EC2 service status
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Open events
None
Upcoming events
None
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Subnets (3) Info
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12. What I learned

- A peering connection alone does nothing; routes on both sides and security group rules decide whether traffic flows.
- CIDR planning must happen first, because overlapping VPC ranges cannot be peered.
- A bastion host with SSH ProxyJump lets me administer private servers without copying keys or giving them public IPs.
- Testing before and after a change, and breaking something on purpose, proves that each rule is actually doing its job.
- Cleaning up promptly avoids unnecessary cost.